Functional Versus Imperative Is Solved

Power, Cost, and Shared State in the Paradigm Debate

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1 Two Ways to Sum a List

Here is the same function written twice.

Listing 1: Imperative, in C.

```c
int total(const int *xs, int n) {
  int acc = 0;
  for (int i = 0; i < n; i++)
    acc += xs[i];
  return acc;
}
```

Listing 2: Functional, in Haskell.

```haskell
total :: [Int] -> Int
total []       = 0
total (x : xs) = x + total xs
```

The first is a list of instructions. The second is a pair of equations. Programmers have argued about which is better for most of the history of programming, and the argument is older than most of the languages it is now conducted in. We settle it.

Definition 1.1 (Imperative)**.** A program is *imperative* if its meaning is a sequence of commands, each of which changes a *store*, a map from locations to values. Running it executes the commands in order, and its result is read from the final store.

Definition 1.2 (Functional)**.** A program is *functional* if its meaning is an expression, and the meaning of an expression is its value. It is *pure* if the value of every expression depends only on the values of its subexpressions, so that any subexpression can be replaced by its value without changing the meaning of the whole. Strachey, borrowing from Quine, called this property *referential transparency* [36].

Remark 1.3. The term has since been used in several senses that are not equivalent [34]. We use Strachey's. Both definitions classify programs, not languages. C permits a function that assigns nothing, and Haskell permits one that does nothing but assign. Most languages in use permit both, so the question is not which language to use but which kind of program to write.

2 The Ceasefire of 1937

The first question, whether one style can compute something the other cannot, was settled before either had a machine to run on. Church proposed that the effectively calculable functions are those definable in the λ-calculus [6]. Turing proposed those computable by a machine that reads and writes symbols on a tape, following a table of instructions [38]. The first is a functional language and the second an imperative one.

Theorem 2.1 (The Church–Turing Ceasefire)**.** *A function on the natural numbers is λ-definable if and only if it is computable by a Turing machine.*

Proof. Turing showed that every λ-definable function is computable and every computable function is general recursive [39], and Kleene that every general recursive function is λ-definable [20]. □

Corollary 2.2 (No Exclusive Programs)**.** *Any function computed by a program in one Turing-complete language, functional or imperative, is computed by some program in any other.*

Proof. Each such language computes at least the functions that Turing machines compute, since it is Turing-complete, and at most these, since a Turing machine can run its interpreter. Theorem 2.1 is the case of the λ-calculus and Turing's machines. □

Remark 2.3. The ceasefire was signed in 1937, eleven years before the first stored-program computer ran a program [41]. It concerns which functions can be computed, not how quickly, how clearly, or how often incorrectly. The rest of this paper is about those.

3 Every Program Is Both

The ceasefire is not only a theorem. Compilers carry it out, in both directions, on most programs that are ever run.

Proposition 3.1 (Lambda, the Ultimate Goto)**.** *A call in tail position, whose result the caller returns unchanged, can be compiled as a jump that reuses the caller's stack frame. A tail-recursive function therefore runs as a loop, in constant stack space [35].*

A functional programmer who wants the loop of Listing 1 writes Listing 3, which GHC compiles to a loop.

Listing 3: Listing 1, as a functional programmer writes it.

```
total :: [Int] -> Int
total xs = go xs 0
  where
    go []       acc = acc
    go (y : ys) acc = go ys (acc + y)
```

Proposition 3.2 (SSA Is Functional Programming)**.** *In* static single assignment *(SSA) form every variable is assigned exactly once, and where control flow merges, a ϕ-function selects a value according to the block that control came from [8]. A program in SSA form can be read as a functional program: each block is a function, each ϕ-function a parameter, and each jump a tail call [17, 2].*

Listing 4: Listing 1 in SSA form.

```
entry:
  br loop
loop:
  i   = phi [0, entry], [i1, body]
  acc = phi [0, entry], [acc1, body]
  br i < n, body, exit
body:
  acc1 = acc + xs[i]
  i1   = i + 1
  br loop
exit:
  ret acc
```

Listing 5: Listing 4, read as functions.

```
entry()      = loop(0, 0)
loop(i, acc) = if i < n then body(i, acc)
               else exit(acc)
body(i, acc) = loop(i + 1, acc + xs[i])
exit(acc)    = acc
```

Clang compiles C to LLVM IR, which is in SSA form throughout [21], and the `mem2reg` pass promotes local variables whose addresses are never taken out of memory and into SSA form [23]. GCC has optimised in SSA form since version 4.0, in 2005 [12]. Arrays and other memory remain loads and stores, so Appel's reading covers the scalar variables [2], but the loop of Listing 1 spends most of its time in the compiler as the functions of Listing 5. C programmers have been writing functional programs for decades. The compiler was too polite to mention it.

The traffic runs the other way too. GHC translates Haskell through two functional intermediate languages, Core and STG, into Cmm, "a low-level imperative language with an explicit stack", and from there into machine code, directly or through LLVM [25]. Lazy evaluation, which in the words of Haskell's designers "kept us pure" [15], is itself implemented by mutation. An argument is passed unevaluated, as a *thunk*, and the first time its value is needed the thunk is overwritten in place with that value, so that it is computed at most once. Wadsworth proposed this in 1971 [44], and GHC's STG machine works this way [30].

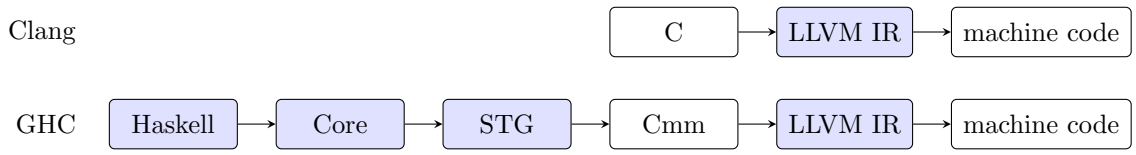


Figure 1: The stages of two compilers, with functional stages shaded: Clang, and GHC with its LLVM backend. Reading SSA form as functional (Proposition 3.2), the C program changes paradigm twice and the Haskell program three times.

Remark 3.3. The paradigm of a program is therefore a property of the stage at which a person reads it. Most of the debate concerns the first stage.

4 Equally Powerful, Not Equally Expressive

Corollary 2.2 allows a translation to rewrite the whole program, which is how compilers use it. Programmers would rather not. Felleisen asked instead which features can be removed by *macro* translation, rewriting each use of a feature from its parts without touching the rest of the program [10]. His test is observational: if a new feature lets some program tell apart two expressions that no program could tell apart before, no macro can remove it.

Theorem 4.1 (Felleisen)**.** *In Pure Scheme, a call-by-value functional language, the expressions* `((lambda (d) (f 0)) (f 0))` *and* `(f 0)`*, which call* `f` *twice and once, are operationally equivalent. Assignment, through* `set!` *and* `letrec`*, separates them, so Pure Scheme cannot macro-express assignment [10].*

Sketch. Without assignment, a function called twice on the same argument does the same thing both times, so the first call can matter only by never returning, and then the single call never returns either. With assignment, `f` can redefine itself on its first call so that its second call never returns. □

Felleisen proves the same for first-class continuations. The theorem's witness is a familiar maintenance problem. Suppose a function deep inside a pure program must now count how often it is called, which is exactly the difference between calling it twice and calling it once. In C this is one global variable and one increment. In Haskell the function's type changes, from `a -> b` to `a -> State Int b`, and so does the type of every function that calls it, directly or not. The imperative programmer pays the same price elsewhere. Keeping the previous version of a data structure, to undo an edit or to read it from another thread, is free in a pure program, where old versions are never destroyed, and needs a copy before every update, or a redesigned structure [9], in an imperative one. Table 1 collects these and two more.

Change	Functional	Imperative
Count the calls to a deep function	every caller	one variable
Keep the previous version of a structure	nothing	every update
Add an operation on a data type	one function	every class
Add a case to a data type	every function	one class

Table 1: What a change touches in each style. The last two rows assume the object-oriented style common in imperative languages, and are the Expression Problem.

Wadler named the last two rows the *Expression Problem*: to define a data type by cases, "where one can add new cases to the datatype and new functions over the datatype, without recompiling existing code, and while retaining static type safety" [43]. Solutions exist, and rely on type-system features developed for the purpose [37].

Remark 4.2 (Conservation of Inconvenience)*.* For changes that cross function boundaries, each style makes some local by making others global. Felleisen's measure ranks languages, not styles, and it rewards having everything: any feature that no macro can remove makes a language more expressive, which may explain C++.

5 The Price of Purity

A pure program cannot update memory in place. Where C overwrites one element of an array, a pure program builds a new array, or a new path through a tree that shares everything else with the old one. Pippenger asked what this costs, in a model of Lisp in which purity means giving up `rplaca` and `rplacd`, the operations that overwrite a cons cell [31]. His programs are *on-line*, producing each output before reading the next input, and *symbolic*, comparing input symbols only for equality.

Proposition 5.1 (The Logarithm Bound)**.** *An impure symbolic on-line program that performs T operations can be simulated by a pure one that performs $O(T \log T)$ [31].*

Proof. Keep the cells that the impure program overwrites in a balanced tree, keyed by when they were created, and copy the path to a cell when it is updated. With at most T cells the tree has depth $O(\log T)$, so each access costs $O(\log T)$, and each new version of the tree shares all but one path with the old. □

Theorem 5.2 (Pippenger)**.** *There is a symbolic on-line problem that an impure Lisp program solves with $O(n)$ operations, while every pure Lisp program needs $\Omega(n \log n)$ operations on some inputs [31].*

The logarithm is therefore sometimes charged, in a model that evaluates everything as soon as it can. Lazy evaluation changes the answer.

Theorem 5.3 (Bird, Jones and de Moor)**.** *With lazy evaluation, a pure program solves Pippenger's problem in linear time overall [5].*

Remark 5.4 (The Refund)*.* Laziness recovers the logarithm through the mechanism of Section 3: an evaluated thunk is overwritten with its value. Bird, Jones and de Moor call this "essentially a restricted memoization", and the capsule review printed with their paper calls it "this restricted form of mutation". They "do not, and cannot, claim that a lazy implementation can solve all problems with the same efficiency as an impure Lisp solution" [5], and as far as we know the question is still open. Purity costs at most a logarithm, and on the problem that shows the logarithm is ever charged, laziness refunds it.

Many structures pay nothing (Table 2). Hood and Melville gave a purely functional queue whose every operation takes constant time in the worst case [14]. Okasaki's simpler one relies on laziness, and he concedes that "some might argue that our implementation is not, in fact, purely functional because of the imperative nature of memoization" [27]. Persistence, the ability to read old versions after an update, comes free with purity, and costs an imperative linked structure of bounded in-degree $O(1)$ amortised time and space per update [9].

Operation	Imperative	Purely functional
Read or write an array element	$O(1)$	$O(\log n)$
Add to or remove from a queue	$O(1)$	$O(1)$ [14]
Sort n items by comparison	$O(n \log n)$	$O(n \log n)$
Keep every old version	$O(1)$ more per update [9]	free

Table 2: Worst-case costs, except for persistence, which is amortised. The purely functional array is a balanced tree, and the persistent structure is linked, with bounded in-degree.

6 Share or Mutate

The strongest argument for functional programming concerns concurrency.

Definition 6.1 (Data race)**.** A *data race* is a pair of accesses to the same memory location by different threads, at least one of them a write, that are not ordered by synchronisation.

Proposition 6.2 (Share or Mutate)**.** *A data race needs a location that is both* shared*, accessed by two threads, and* mutable*, written by at least one of them. A program in which no location is both has no data races.*

Proof. Definition 6.1. □

Concurrency bugs in practice look like this. Of 105 sampled from MySQL, Apache, Mozilla and OpenOffice, 74 did not involve deadlock, and 72 of those were atomicity or order violations, in which accesses to memory shared between threads happened in an order the programmer had not intended [24].

A pure program writes nothing once a value is built, so it has no data races, and its parts can be evaluated in any order, or at once, without changing its result. For the λ-calculus this is a theorem.

Theorem 6.3 (Church–Rosser)**.** *If a λ-term reduces to two terms, then both reduce to a common term. In particular a term has at most one normal form, whatever order its reductions are performed in [7].*

Permission to evaluate in parallel is not a speedup. If a fraction f of a program's work can be spread over p processors and the rest cannot, the speedup is at most

$$S(p) = \frac{1}{(1-f) + f/p} < \frac{1}{1-f}, \tag{1}$$

a bound now called Amdahl's law after his argument in [1]. A program that is 90% parallel runs at most ten times faster, on any number of processors, in either style.

Imperative languages can remove the sharing instead of the mutation. In Rust, "at any given time, you can have either one mutable reference or any number of immutable references", and because the compiler enforces this, "Rust can prevent data races at compile time" [18]. RustBelt proved the discipline sound for a formal model of the language, including libraries that use unsafe code inside: a well-typed program "will not have data races" [16].

Pure languages, from the other side, have learned to mutate what is not shared. Values of a linear type "must be used exactly once", and so "safely admit destructive array update" [42]. GHC has supported linear types since version 9.0 [4, 13]. Perceus, the reference counting of the Koka language, checks at run time whether a reference is unique, "to update in-place when possible" [33].

Remark (The Paradigm War, Stated Precisely). Proposition 6.2 is the rigorous content of the debate. The hazard was never mutation, and never sharing, but the two at once. Functional programming forbids mutation, and pays for it in Section 5. Rust forbids sharing while mutating, and pays in annotations and a borrow checker. Linear types let pure programs mutate what is unshared, and ownership lets imperative programs share what is immutable, so the two paradigms have reached the same rule from opposite ends.

Share or mutate. Not both.

7 What the Evidence Says

Theorems settle what is possible. Whether either style produces better programs in practice is an empirical question, and the evidence is thinner than the debate.

The largest study of defects, by Ray et al., modelled bug-fixing commits in 729 GitHub projects written in 17 languages, with 80 million lines of code and 1.5 million commits. It found "a small but significant relationship between language class and defects", with functional languages doing better than procedural or scripting ones. Language accounted for less than one percent of the deviance, and the authors warned that even this might reflect "the preference of certain personality types for functional, static and strongly typed languages" [32]. A reproduction by Berger et al. found flaws in the data and the analysis, including a false-positive rate of 36% among the commits labelled as bug fixes. After correcting them, the number of languages associated with defects fell from eleven to four: C++ with more, and Clojure, Haskell and Ruby with fewer, all by amounts the authors call "exceedingly small". Causation, they add, "is not supported by the data at hand" [3]. A Bayesian reanalysis of the same data concluded that the question "does not admit a simple, generally valid answer" [11].

Controlled experiments are smaller, and they split. Uesbeck et al. had 54 programmers, from first-year students to professionals, solve tasks in C++ with either lambdas or an iterator class. The lambda group took longer and made about twice as many compiler errors, and there was no benefit from lambdas at any level of experience, professionals included, though the authors suspect a novelty effect [40]. Mehlhorn and Hanenberg had 20 programmers read Java code that processed collections with loops or with streams. With loops they needed 78% more time and gave three times as many wrong answers [26]. One study measured writing and the other reading, and the functional style lost the first and won the second. Pankratius et al. had 13 programmers write the same projects in both Scala and Java. The Scala code was shorter, but the study refuted Scala's claims of lower programming and debugging effort [28].

Speed and energy depend on the implementation more than on the style. Pereira et al. ran programs for the same benchmark problems in 27 languages and measured their energy, time and memory [29]. Averaged over the authors' classes of paradigm, imperative languages used 125 J and functional ones 1,367 J. The classes overlap, however, and the functional class, which includes Rust and Perl, spans a factor of 77, from 1.03 times the energy of C to 79.58 (Figure 2). Of that class's total, 90% comes from Erlang, Ruby and Perl, which run on a virtual machine or an interpreter. The compiled functional languages OCaml and Haskell used 2.40 and 3.10 times the energy of C, close to Go at 3.23 and C# at 3.14.

Two landmark verified systems sit on the border. CompCert is a C compiler programmed in Coq, which serves as both its programming language and the proof assistant that verifies it, and extracted to Caml [22]. Testing compilers with random C programs, Yang et al. reported more than 325 previously unknown bugs, and of CompCert wrote that "the middle-end bugs we found in all other compilers are absent". As of early 2011 it was "the only compiler we have tested for which Csmith cannot find wrong-code errors"; the bugs they did find were in its unverified front end [45]. The seL4 kernel is 8,700 lines of C and 600 of assembler, proved correct with 200,000 lines of proof script in Isabelle/HOL. Its designers wrote it first as a Haskell prototype, translated that into the prover as an executable specification, and then reimplemented it by hand in C [19]. CompCert is a functional program that compiles imperative ones, and seL4 an imperative program designed in a functional language.

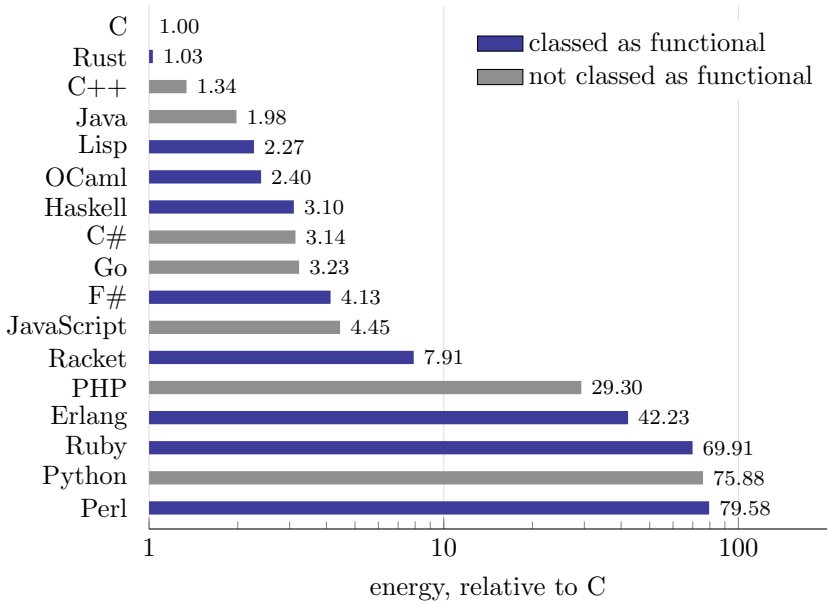


Figure 2: Energy used by the benchmark programs of Pereira et al. [29], relative to C, for every language their table of paradigms classes as functional and eight that it does not. The classes overlap: Rust, OCaml and F# are also classed as imperative, and Perl and Ruby as scripting.

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